

## **EXPERIMENT 5**

**Aim :** To Build the pipeline of jobs using Maven / Gradle / Ant in Jenkins, create a pipeline script to Test and deploy an application over the tomcat server

### **Theory :**

Continuous Integration and Continuous Deployment (CI/CD) is a software development practice that automates the process of building, testing, and deploying applications. Jenkins is a widely used open-source automation server that enables developers to create pipelines for integrating code changes, running tests, and deploying applications efficiently. In this experiment, build tools such as Apache Maven, Gradle, or Apache Ant are used to compile source code, manage dependencies, and package the application.

A Jenkins pipeline is defined using a script (commonly called a Jenkinsfile), which specifies stages such as Build, Test, and Deploy. During the build stage, the selected tool (Maven/Gradle/Ant) compiles the project and generates an artifact (e.g., WAR file for web applications). In the testing stage, automated test cases are executed to ensure code quality. Finally, in the deployment stage, the generated artifact is deployed to a web server such as Apache Tomcat, which hosts and runs Java-based web applications.

This pipeline-based approach ensures faster development cycles, early detection of errors, and consistent deployment practices. By integrating build tools with Jenkins and deploying on Tomcat, organizations achieve automation, scalability, and reliability in their software delivery process.

Status

</> Changes

Console Output

Edit Build Information

Delete build '#1'

Timings

✓ #1 (Feb 3, 2026, 2:21:11 PM)

🕒

Started by anonymous user

🕒

This run spent:

- 0 ms waiting;
- 0.11 sec build duration;
- 0.11 sec total from scheduled to completion.

</>

No changes.

New Item - Jenkins

localhost:8080/view/all/newJob

Jenkins / All / New Item

New Item

Enter an item name

Select an item type

Pipeline

Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.

Freestyle project

Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.

Multi-configuration project

Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.

Folder

Creates a container that stores nested items in it. Useful for grouping things together. Unlike view, which is just a filter, a folder creates a separate namespace, so you can have multiple things of the same name as long as they are in different folders.

Multibranch Pipeline

Creates a set of Pipeline projects according to detected branches in one SCM repository.

Organization Folder

Creates a set of multibranch project subfolders by scanning for repositories.

If you want to create a new item from other existing, you can use this option:

Copy from

OK

Configure

General

Source Code Management

Triggers

Environment

Build Steps

Post-build Actions

☐ Poll SCM ?

Environment

Configure settings and variables that define the context in which your build runs, like credentials, paths, and global parameters.

- ☐ Delete workspace before build starts
- ☐ Use secret text(s) or file(s) ?
- ☐ Add timestamps to the Console Output
- ☐ Inspect build log for published build scans
- ☐ Terminate a build if it's stuck
- ☐ With Ant ?

Build Steps

Automate your build process with ordered tasks like code compilation, testing, and deployment.

Execute Windows batch command ?

Command

See the list of available environment variables

echo Hello !

Advanced ▾

+ Add build step

Save Apply



**Conclusion :** Successfully implemented the pipeline of jobs using Maven / Gradle / Ant in Jenkins, create a pipeline script to Test and deploy an application over the tomcat server.